

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I P. Guru/192521010 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Guru

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Short Answers Test:

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Ans:

Cloud computing:

Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking and software over/through the internet. It allows the users to pay for the particular resources that they use. It avoids unnecessary payments and wastage of the resources.

Characteristics:

On-Demand self-service:

- Users can access computing resources whenever needed without human intervention and interruption.

Measured service:

- The users only pay for the resources they use.

Resource Pooling:

- Cloud provides share computing resources among multiple users using a multi-tenant model.

Broad network access:

- Services are available over the internet and can be accessed using laptops, smartphones, etc.

2) Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Ans:

Evolution of cloud computing:

1) Hardware evolution:

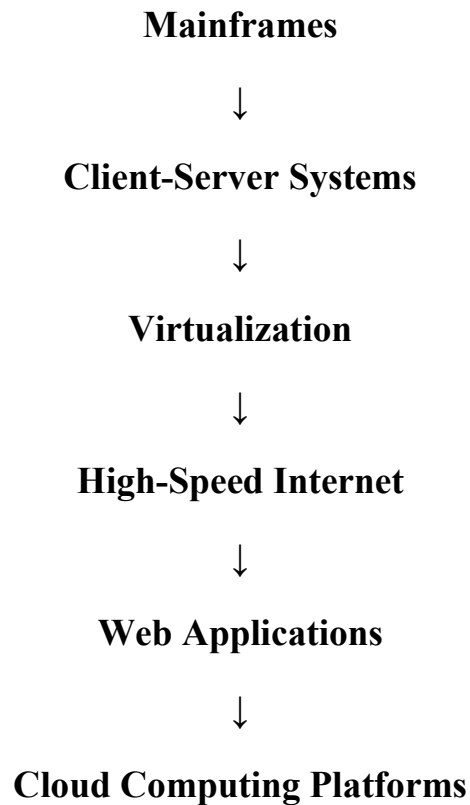
- Early computers were large mainframes.
- Client-server architecture enabled multiple users.
- Powerful servers and virtualization improved hardware utilization.

Internet Evolution:

- High-speed internet enabled remote access to computing resources.
- Broadband and fiber networks reduced latency and increased reliability.

Software Evolution:

- Applications evolved from desktop software to web-based applications.
- Technologies like virtualization, distributed computing, and web services enabled cloud platforms.

Flowchart:

3) Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Ans: Role of Server Virtualization in Cloud Computing:

Server Virtualization:

- Server virtualization is the process of creating multiple virtual machines (VMs) on a single physical server using a hypervisor.

Role in Cloud Computing:

- Improves hardware utilization.
- Reduces infrastructure cost.
- Enables rapid deployment of virtual machines.
- Provides scalability and resource isolation.

Difference:

1)Virtualization:

- Technology
- Creates virtual machines
- Focuses on efficient hardware usage
- Implementation using hypervisors

2)Cloud computing:

- Service model
- Creates virtual machines
- Delivers services over the internet
- Focuses on efficient hardware usage
- Focuses on providing computing services
- Implemented using hypervisors
- Uses virtualization along with networking and automation

4)SOAP vs REST Web Servers:

SOAP:

- Protocol
- Uses XML only
- More complex
- Uses HTTP, SMTP, TCP
- Higher overhead
- Better security and reliability

REST:

- Architectural style
- Supports JSON, XML, HTML, Etc.
- Simple and lightweight
- Mainly uses HTTP
- Faster performance
- Better for web and mobile applications

Therefore, SOAP is preferred to secure enterprise applications

REST is widely used in cloud computing because it is lightweight, scalable, and efficient.

5)Differentiate IaaS, PaaS, SaaS and XaaS.

IaaS: Provides virtual servers, storage, and networking.

Eg: Amazon EC2

PaaS: Provides a platform for application for application development and deployment.

Eg: Google app engine

SaaS: Delivers ready to use software over the internet.

Eg: Google docs.

XaaS : Any IT services delivered over the cloud, such as storage, database, AI, or security as a service.

Eg:Microsoft azure

